Econ 711 hw 2

variances are expectations and conditional on X in all parts below.

1. a OLS estimator, can be written as $(X^TX)^{-1}X^Ty$, while GLS estimator can be written as $(\hat{X}^T\hat{X})^{-1}\hat{X}^T\hat{y}$, in which $\hat{X} = \Sigma^{-\frac{1}{2}}X$, $\hat{y} = \Sigma^{-\frac{1}{2}}y$, while $\Sigma = Var(u|X)$

$\therefore \hat{\beta}_{GLS} = (X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1}y$. Then we can let linear map $A = (X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1}$, and $B = (X^TX)^{-1}X^T$, we have $b = By$, $\hat{\beta}_{GLS} = Ay$

Then, $Cov(\hat{\beta}_{GLS}, b-\hat{\beta}_{GLS}) = Cov(Ay, (B-A)y) = A\cdot Cov(y,y)(B-A)^T = A\cdot Var(y)\cdot(B-A)^T$

Considering $Var(y) = Var(X\beta+u)$, in which β is the true, non-random parameter, and u is unobservable, $Var(y) = \mathbb{E}((y-\underbrace{\mathbb{E}(y)}_{=\mathbb{E}(X\beta+u) = X\beta+\mathbb{E}(u) = X\beta})(y-\mathbb{E}(y))^T) = \mathbb{E}(uu^T) = Var(u) = \Sigma$

Then, $Cov(\hat{\beta}_{GLS}, b-\hat{\beta}_{GLS})$

$= A\Sigma B^T - A\Sigma A^T$

as $A\Sigma B^T = ((X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1})\Sigma\cdot((X^TX)^{-1}X^T)^T = (X^T\Sigma^{-1}X)^{-1}X^TX\cdot[(X^TX)^{-1}]^T = (X^T\Sigma^{-1}X)^{-1}$

$A\Sigma A^T = ((X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1})\Sigma((X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1})^T = (X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1}X\cdot[(X^T\Sigma^{-1}X)^{-1}]^T$

$= (X^T\Sigma^{-1}X)^{-1}$, as $A\Sigma B^T = A\Sigma A^T$, $Cov(\hat{\beta}_{GLS}, b-\hat{\beta}_{GLS}) = 0$

(note that Σ^{-1}, $(X^TX)^{-1}$ and $(X^T\Sigma^{-1}X)^{-1}$ are symmetric, hence equal to their transpose)

b. Since $\hat{\beta}_{GLS}$ and $\check{\beta}$ are unbiased estimators, $q = \check{\beta} - \hat{\beta}_{GLS}$, we have $\mathbb{E}(q) = 0$

Then, let H be any fixed matrix, we can have a family of unbiased estimators $\hat{\beta}(H) := \hat{\beta}_{GLS} + Hq$, since $\mathbb{E}(q) = 0$, we have that $\mathbb{E}(\hat{\beta}(H)) = \beta$ $\therefore$ unbiased. Then, we can compute the variance of $\hat{\beta}(H)$:

$Var(\hat{\beta}(H)) = Var(\hat{\beta}_{GLS}+Hq) = Var(\hat{\beta}_{GLS}) + \underbrace{Var(Hq)}_{=H\cdot Var(q)\cdot H^T} + H\,Cov(q, \hat{\beta}_{GLS}) + Cov(q, \hat{\beta}_{GLS})\cdot H^T$

$= Var(\hat{\beta}_{GLS}) + \underbrace{HC^T + CH^T}_{\text{by letting } C := Cov(q, \hat{\beta}_{GLS})} + H\,Var(q)H^T$

Then, if $Cov(q, \hat{\beta}_{GLS}) \neq 0$, $C \neq 0$, by letting $H^* = -C(Var(q))^{-1}$

we have $Var(\hat{\beta}(H^*)) = Var(\hat{\beta}_{GLS}) - C\underbrace{(Var(q))^{-1}}C^T$

$\therefore$

Since $C \neq 0$, $C(Var(q))^{-1}C^T$ is P.S.D. and non-zero and $(Var(q)^{-1})$ is P.D.

Thus, we have that $Var(\hat{\beta}_{GLS}) \succ Var(\hat{\beta}(H^{\circ}))$

$\therefore$ contradiction to the efficiency of GLS estimator!

Hence, we must have $C = 0$, $Cov(q, \hat{\beta}_{GLS}) = 0$

c. Since V is an $n \times n$ variance-covariance matrix of ε,

we know that V is a symmetric matrix. hence $\exists$ orthogonal matrix that

$H^T H = I_n$, and $H^T V H = \Lambda$, in which Λ is a diagonal matrix with the
(orthogonal eigenvalues)

characteristic roots of V (eigen decomposition) in the diagonal
(eigenvalues)

and columns of H are chara. vectors (eigenvectors) of V. Then, we have

$H \Lambda H^T = V$, $\quad H \Lambda^{-1} H^T = V^{-1}$

(all K)
if the columns of X are eigenvectors of V, W.L.O.G. let them be first K cols of

H, and hence, let $\underset{(n \times k)}{F} = \begin{pmatrix} I_k \\ 0 \end{pmatrix} \begin{matrix} \}(k \times k) \\ \}((n-k) \times k) \end{matrix}$, we have $X = HF$.

"selector matrix

Then, OLS estimator b can be written as:

$$(X^T X)^{-1} X^T y = (\underbrace{F^T H^T H F}_{= F^T I_n F = I_k})^{-1} X^T y = X^T y = F^T H^T y$$

GLS estimator $\hat{\beta}_{GLS}$ can be written as:

$$(X^T V^{-1} X)^{-1} X^T V^{-1} y = (\underbrace{F^T H^T V^{-1} H F}) ^{-1} X^T V^{-1} y = \Lambda_k X^T V^{-1} y = \underbrace{\Lambda_k F^T H^T V^{-1}} y$$

as $H \Lambda^{-1} H^T = V^{-1}$, $H^T V^{-1} H = \Lambda^{-1}$, $F^T \Lambda^{-1} F$
includes the upper-left $k \times k$ block of Λ^{-1}, calling it Λ_k^{-1}

$$\therefore = \Lambda_k F^T H^T (H \Lambda^{-1} H^T) y = \Lambda_k F^T \Lambda^{-1} H^T y$$

since F^T picks first k diagonal entries of Λ^{-1}, we have $\hat{\beta}_{GLS} = F^T H^T y$

$\therefore \hat{\beta}_{GLS} = b$.